

SUMMARY

1.1 Introduction to Multimedia

1.1.2 Elements of multimedia

- a) Describe multimedia elements.
- b) Explain the purpose of each multimedia elements.

Multimedia Elements	Definition	Purpose
Text	Combination of letters that form a word or sentence that describes a topic which is known as text information.	Text is effective for conveying basic information
Audio	The use of speech, music and sound effects produced by vibration, as perceived by the sense of hearing.	Audio can add an exciting dimension to a presentation
Graphic	A digital representation of non-text information Graphic makes complex information easy to understand	Graphic make complex information easy to understand
Animation	A series of graphical images displayed one after another to simulate movement	Animation is useful for illustrating concepts which involve movement.
Video	A series of frames containing pictures and sound played back at a fast rate	Videos can provide real situation to help users understanding

1.1.3 Applications of Multimedia

b) Differentiate two-dimensional (2D), three-dimensional (3D) and four-dimensional (4D) design.
(Axis, shape, dimension)

Dimension Full Form	2D	3D	4D
	Two-Dimensional	Three-Dimensional	Four-Dimensional
Definition	Represents an object with just two dimensions, i.e., length and height.	Represents an object with three dimensions: Length, width, and height	Represents an object with four dimensions: Length, width, height, and time/motion.
Axis	Use of x-axis and y-axis.	Use of x-axis, y-axis and the z-axis.	Apply x-axis, y-axis, and the z-axis to a space with four dimensions.
File Size	Relatively smaller than 3D and 4D.	Relatively smaller than 4D and larger than 2D.	Relatively larger than 2D and 3D.
Rendering Time	Relatively shorter time.	Relatively longer than 2D but shorter time than 4D.	Take long hours
Shape (Geometry)	• Circles • squares • triangles • any flat surface or image	• Spheres • Cubes • pyramids, tetrahedron • any solid object/life like	Abstract concept, Polychora (4-polytopes) made of polyhedra (made of two dimensional polygons).
Dimension	Length, and height no depth (width).	Length, width, and height	Length, width, height, and time/motion.
Usage	Regular films In education 2D graphics might be used for illustrating concepts on a whiteboard or in presentations	3D Films that use optical illusion to create the representation of a three-dimensional film. In education 3D models could enhance the understanding of complex structures or systems in subjects like biology, chemistry, or physics.	A 3D film with the added effects that take place in specially designed theatres. In education 4D allows interactive learning. It allows students to actively participate in scenarios and learn through exploration, often involving the fourth dimension of time.

Benefits/Advantages	Simplicity: 2D graphics are simpler and easier to create than 3D graphics. They are often more straightforward for artists and designers to work with. Clarity: In certain applications like presentations, maps, or diagrams, 2D graphics can provide clear and easily understandable representations of information. Resource Efficiency: 2D graphics generally require less computational power and resources compared to 3D graphics, making them more suitable for less powerful devices.	Realism: 3D graphics can create a more realistic representation of objects and scenes, making them suitable for applications such as video games, simulations, and movies. Depth Perception: 3D graphics allow for depth perception, providing a more immersive experience. This is particularly beneficial in virtual reality (VR) environments. Versatility: 3D models can be viewed from various angles, offering greater flexibility in terms of presentation and interaction.	Time Dimension: 4D adds the element of time to 3D graphics, allowing for dynamic and time-based changes. This is often used in fields like computer graphics, simulations, and scientific visualization. Simulation and Analysis: In certain applications, such as physics simulations or architectural planning, adding a time dimension (the fourth dimension) can provide insights into how objects or systems evolve over time. Interactive Experiences: 4D can enhance interactive experiences by incorporating time-based elements, creating more engaging and dynamic content.
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1.2 Multimedia Design

1.2.1 Three Phase Multimedia Design

b) Describe pre-production, production, and post-production phases.

Pre-production	Production	Post-production
The process before producing multimedia project.	The process in which multimedia project is produced.	The process after the production of multimedia project.
1. Planning – determine the content, time taken, the cost and person in charge of the project	1. Digitizing Arts - Converting existing assets into digital for the use in an electronic medium	1. Testing - The product is tested on multiple computers from different users.
2. Resource Organization - The content is arranged categories so that comprehensive task planning can be executed.	2. Modelling & Animation - Models of artwork are created, rendered, and then prepared for use in the authoring tool	2. Mastering - Writing the final multimedia products in suitable platform.
3. Flowchart – a visual representation to outline the content.	3. Shooting Video - Shoot video based on the edited scripts	3. Archiving - The original multimedia files are archived for future upgrade and revisions.
4. Navigation - the signpost of each task by using CPM, PERT chart or Gantt chart	4. Recording Audio - Composed audio to be used with other multimedia elements	4. Duplication - The duplicates are created from the original and packaged accordingly
5. Define Screen Action - screen or interface action directs how the interface responds to the user.	5. Authoring - All the pieces come together in the authoring tool. The final working product is created	5. Distribution - The final step in the process is distributing your multimedia project.
	6. Proofreading - Every word on the screen is proofread and checked for consistency.	
	7. Quality control - Quality control goes on throughout the process.	

1.2.2 Multimedia Authoring Tools

- b) Differentiate among types of multimedia authoring tools
- c) Explain multimedia navigation (hypertext, hypermedia, tab, audio, hand gesture)

	TIME-BASED	ICON-BASED	PAGE-BASED
Characteristics	The <u>multimedia elements or events</u> are presented and <u>organized along a timeline</u>	The <u>multimedia elements or events</u> are <u>organized as objects in a structural framework</u>	The <u>multimedia elements or events</u> are <u>organized as pages of a book or a stack of a cards.</u>
Usage	Animation and video	Interactivity	Educational content
Example of software	Adobe Animate, MOHO	Scratch, Icon Author	Google Slide, Prezi

	Multimedia Navigation	Explanation
1	Hypertext	Refers to the system of managing the information related to the plain text. OR Link in text-based document.
2	Hypermedia	Refers to connecting the hypertext with media such as graphics, sounds and animations. OR Link in non-text-based document
3	Tab	Clickable area at the top that shows another page or area.
4	Audio	Refers to control and handling multimedia applications using audio (voice search)
5	Hand Gesture	Control and handling multimedia applications through hand gesture recognition scheme.
6	Menu	The simplest menus consist of text lists of topics. Users choose a topic and click it.

1.3 Multimedia Delivery

1.3.1 Differentiate offline and online Multimedia Delivery.

Criteria	ONLINE	OFFLINE
Accessibility	Require broadband Internet connection	Do not require broadband Internet connection / Can be accessed without internet.
Ease of editing/Content	Can be changed and updated easily.	Permanent and not changeable.
Speed (data transfer)	Subject to the constraints of the viewer's Internet Service Provider (ISP) or network.	Relatively faster data transfer rate.
Capacity	Limited in picture size and low resolution video	Can store high end multimedia elements such as video
Example	Example of delivery: • Web-based • Cloud-based	Example of delivery: • hard disk • solid state drive (SSD) • CD/DVD/Blu-Ray